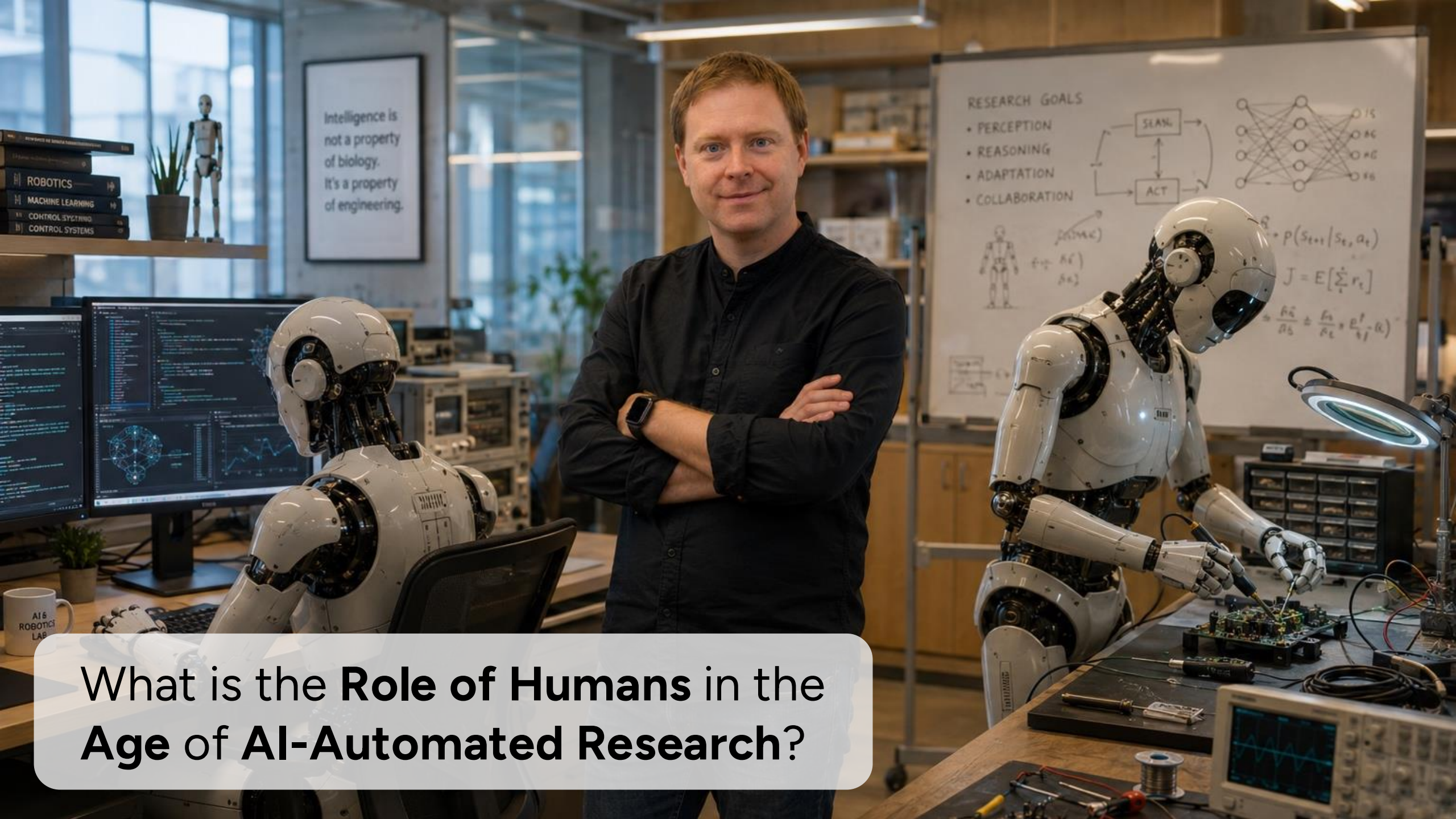
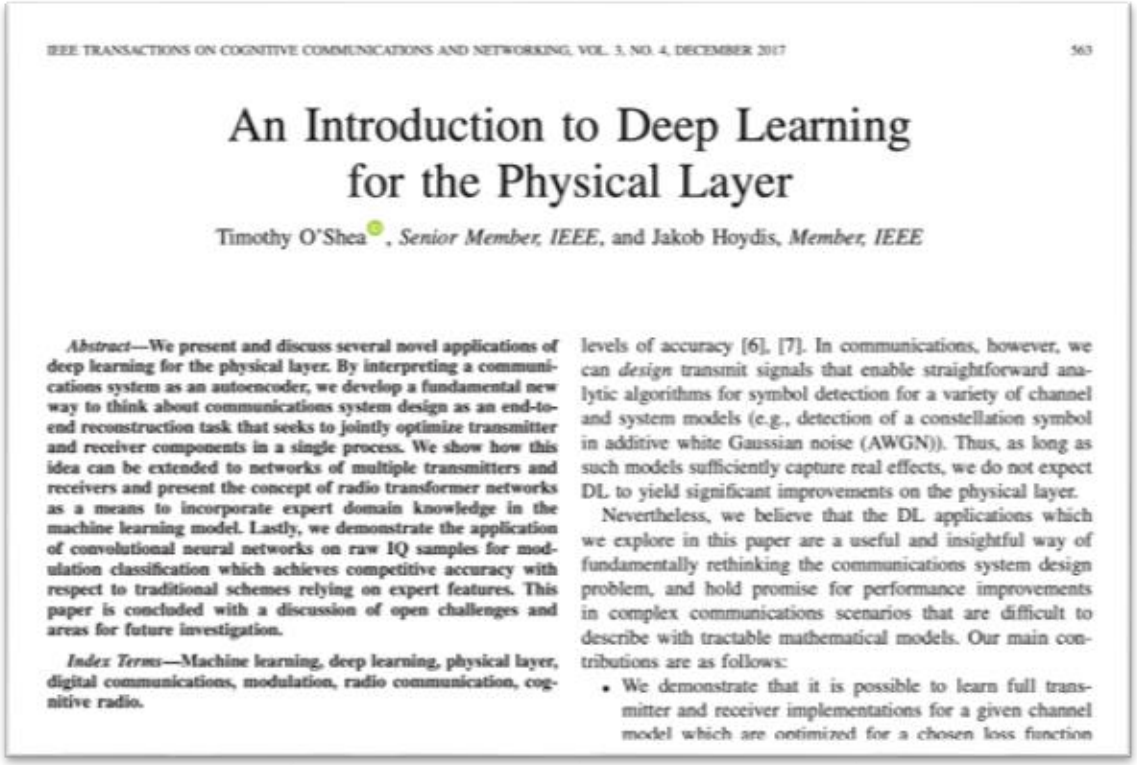


What is the **Role of Humans** in the **Age** of **AI-Automated Research**?

10+ Years of **AI Research** in Communications Community

2017: Seminal paper



IEEE TRANSACTIONS ON COGNITIVE COMMUNICATIONS AND NETWORKING, VOL. 3, NO. 4, DECEMBER 2017 563

An Introduction to Deep Learning for the Physical Layer

Timothy O'Shea, *Senior Member, IEEE*, and Jakob Hoydis, *Member, IEEE*

***Abstract*—We present and discuss several novel applications of deep learning for the physical layer. By interpreting a communications system as an autoencoder, we develop a fundamental new way to think about communications system design as an end-to-end reconstruction task that seeks to jointly optimize transmitter and receiver components in a single process. We show how this idea can be extended to networks of multiple transmitters and receivers and present the concept of radio transformer networks as a means to incorporate expert domain knowledge in the machine learning model. Lastly, we demonstrate the application of convolutional neural networks on raw IQ samples for modulation classification which achieves competitive accuracy with respect to traditional schemes relying on expert features. This paper is concluded with a discussion of open challenges and areas for future investigation.**

***Index Terms*—Machine learning, deep learning, physical layer, digital communications, modulation, radio communication, cognitive radio.**

levels of accuracy [6], [7]. In communications, however, we can *design* transmit signals that enable straightforward analytic algorithms for symbol detection for a variety of channel and system models (e.g., detection of a constellation symbol in additive white Gaussian noise (AWGN)). Thus, as long as such models sufficiently capture real effects, we do not expect DL to yield significant improvements on the physical layer.

Nevertheless, we believe that the DL applications which we explore in this paper are a useful and insightful way of fundamentally rethinking the communications system design problem, and hold promise for performance improvements in complex communications scenarios that are difficult to describe with tractable mathematical models. Our main contributions are as follows:

- We demonstrate that it is possible to learn full transmitter and receiver implementations for a given channel model which are optimized for a chosen loss function

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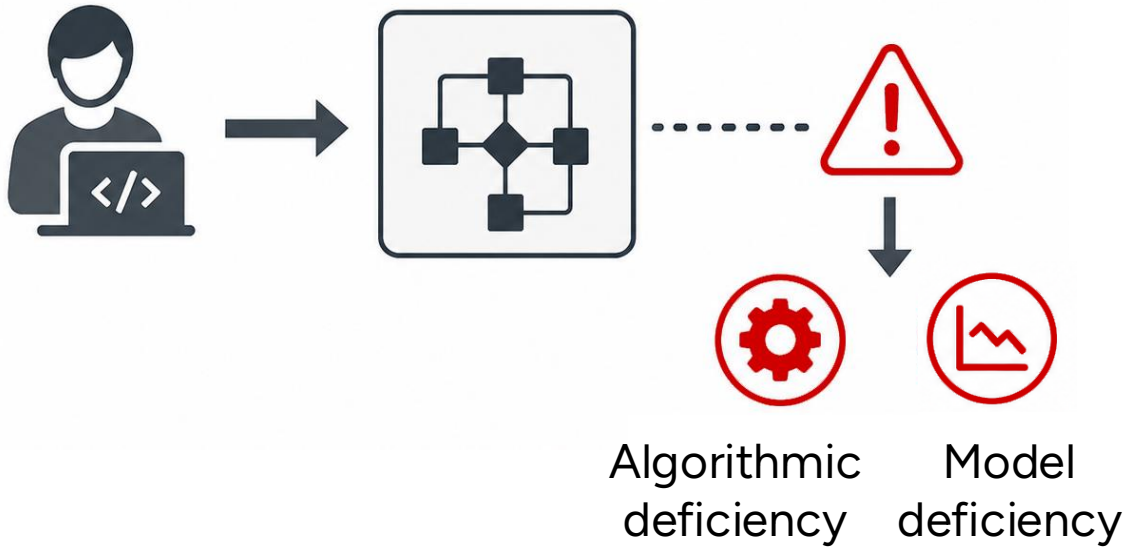
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SAC Machine Learning for Communications and Networking

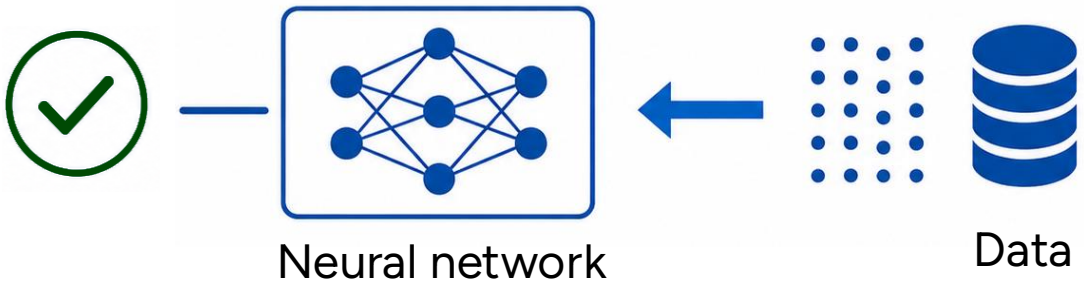
20 sessions
100 papers

1) AI for Communications

Model-driven design

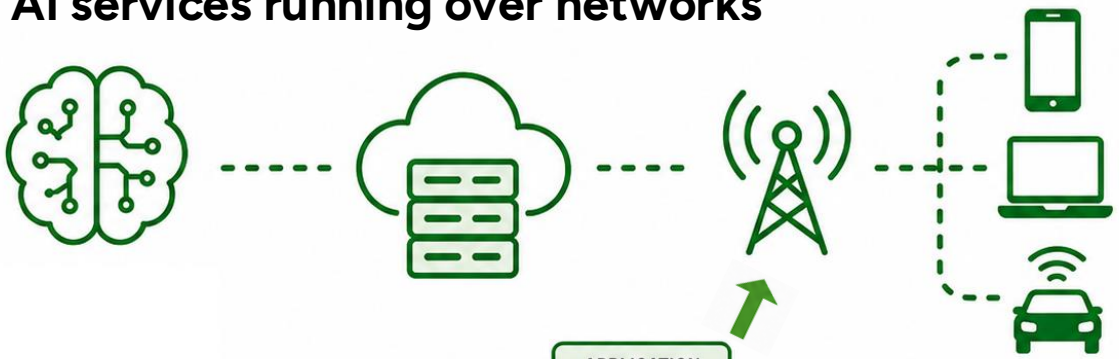


Data-driven design



2) Communications for AI

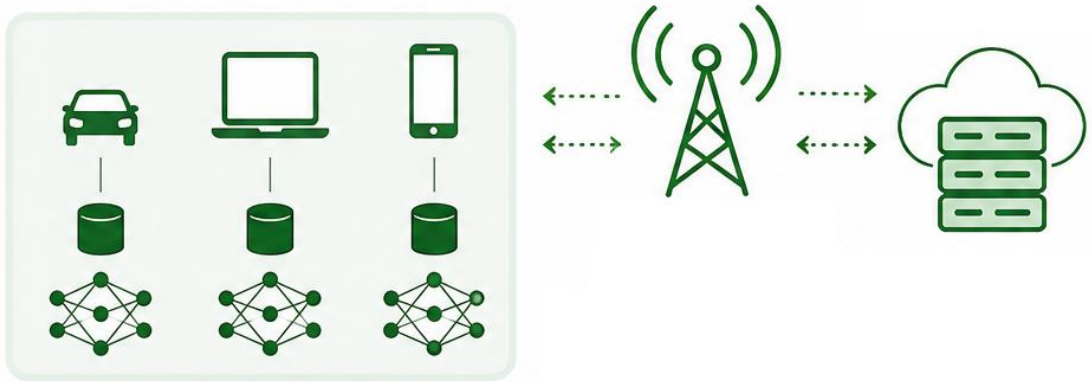
AI services running over networks



Revised protocols



Federated learning

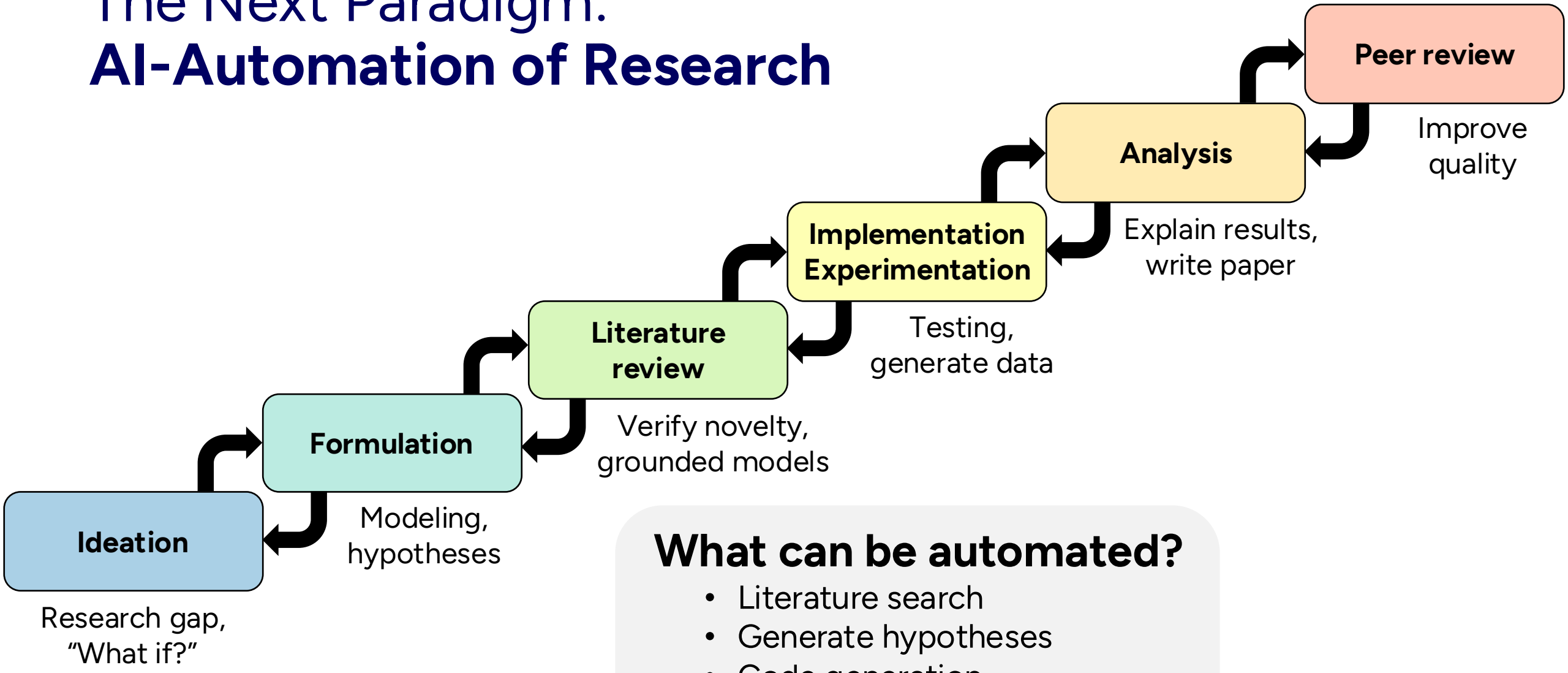




What happens next changes everything...



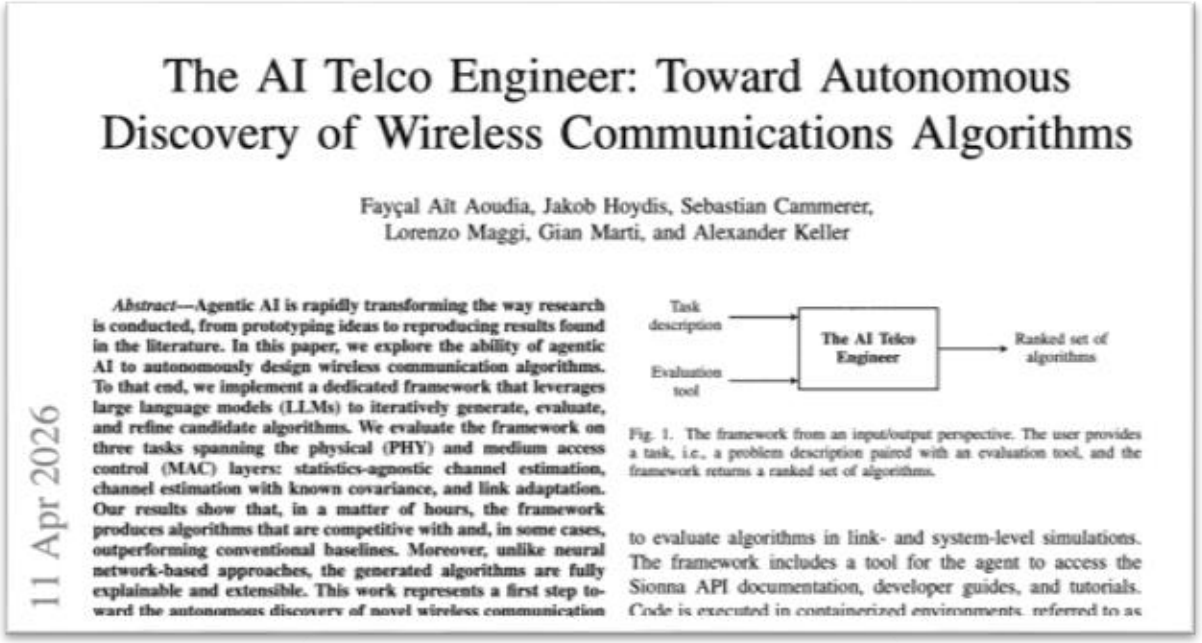
The Next Paradigm:
AI-Automation of Research



What can be automated?

- Literature search
- Generate hypotheses
- Code generation
- Text writing and revision
- Interaction with simulators

The **AI Telco Engineer:** An NVIDIA Framework



The AI Telco Engineer: Toward Autonomous Discovery of Wireless Communications Algorithms

Fayçal Aït Aoudia, Jakob Hoydis, Sebastian Cammerer,
Lorenzo Maggi, Gian Marti, and Alexander Keller

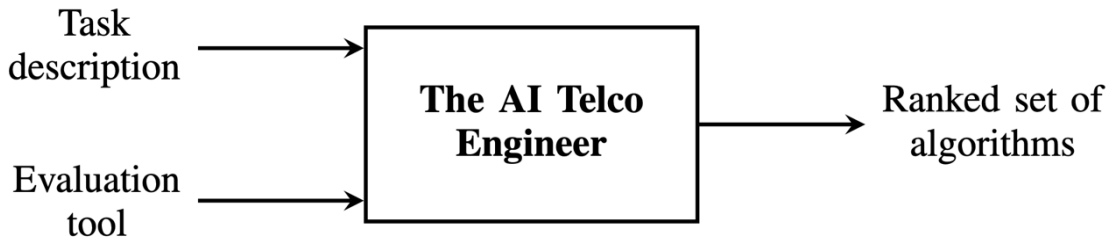
Abstract—Agentic AI is rapidly transforming the way research is conducted, from prototyping ideas to reproducing results found in the literature. In this paper, we explore the ability of agentic AI to autonomously design wireless communication algorithms. To that end, we implement a dedicated framework that leverages large language models (LLMs) to iteratively generate, evaluate, and refine candidate algorithms. We evaluate the framework on three tasks spanning the physical (PHY) and medium access control (MAC) layers: statistics-agnostic channel estimation, channel estimation with known covariance, and link adaptation. Our results show that, in a matter of hours, the framework produces algorithms that are competitive with and, in some cases, outperforming conventional baselines. Moreover, unlike neural network-based approaches, the generated algorithms are fully explainable and extensible. This work represents a first step toward the autonomous discovery of novel wireless communication

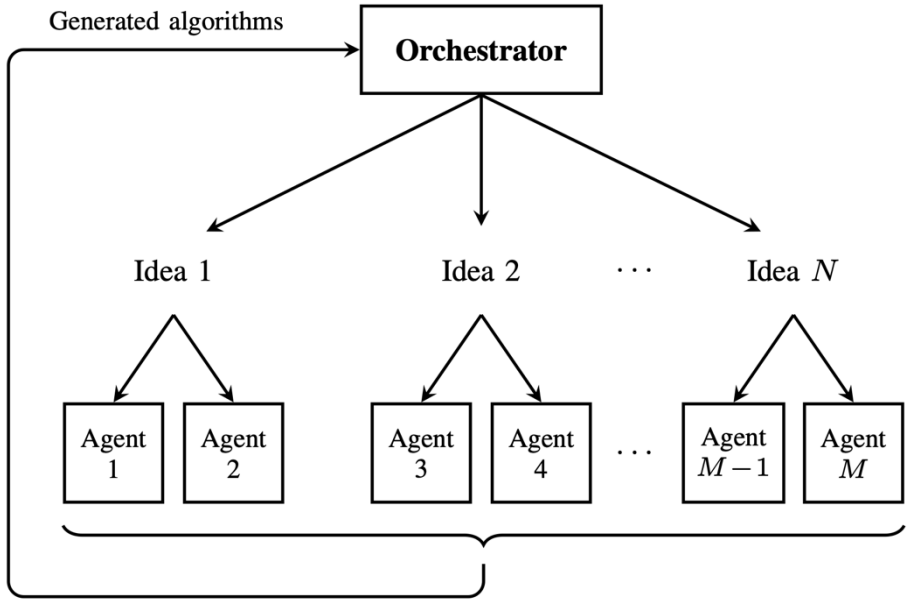
11 Apr 2026

Task description
Evaluation tool
The AI Telco Engineer
Ranked set of algorithms

Fig. 1. The framework from an input/output perspective. The user provides a task, i.e., a problem description paired with an evaluation tool, and the framework returns a ranked set of algorithms.

to evaluate algorithms in link- and system-level simulations. The framework includes a tool for the agent to access the Sionna API documentation, developer guides, and tutorials. Code is executed in containerized environments, referred to as





Generate PHY/MAC algorithms

- LLM, Agentic AI, Sionna simulator
- Competitive performance
- Explainable selection

Four Key Roles of Humans

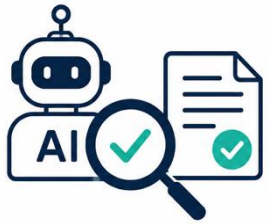
Think outside the box



Problem solvers → Problem finders

Extrapolate beyond current knowledge

Counteract hallucination



Supervise AI agents, verify correctness

Ensure scientific soundness

Select performance metrics



Multi-objective problems → One objective

Technical, economic, vs. ethical constraints

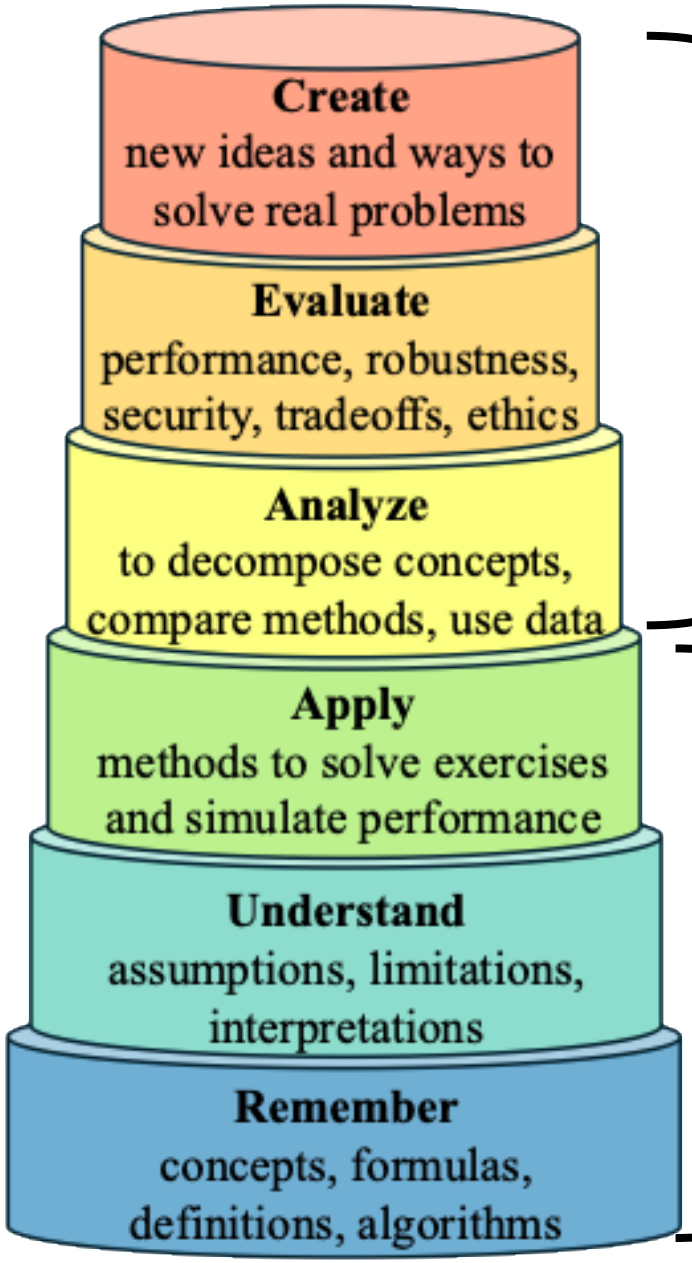
Teamwork in experimentation



Raise technology readiness level

Combine hardware/software in real world

Teaching the Next Generation Humans



Bloom's taxonomy

Core human competencies
Deep thinking, judgments, real system design

*"AI agents are your **paid** assistant"*

Can be deeply automated
Teach deep fundamental understanding

"Learn to count before using calculators"

Implications on the **Scientific Publication System**



Explosion of papers

Raise the quality bar

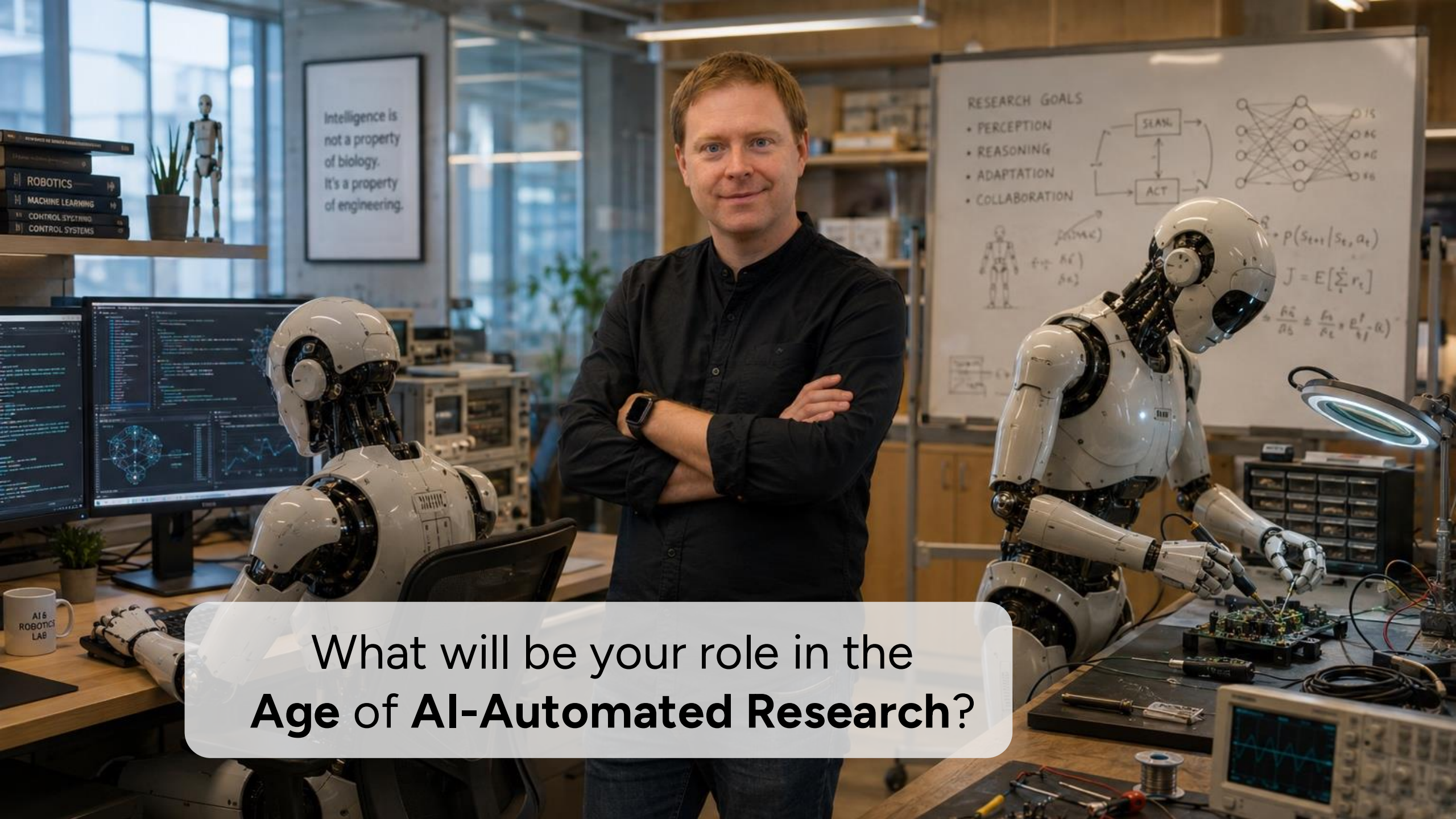


Require reproducibility





Curation of knowledge, scientific debate



What will be your role in the **Age** of **AI-Automated Research**?

ACCEPTED FROM OPEN CALL

Automating Wireless Research and Development: What Remains Human?

Emil Björnson, Mischa Dohler, Jakob Hoydis, and Robert W. Heath Jr.

IEEE COMMUNICATIONS MAGAZINE

Thank you for listening!

More details in our paper:

Abstract

The rapid advancement of artificial intelligence (AI) is fundamentally disrupting research and engineering. While much attention is given to how AI may optimize wireless systems, this article explores a different question: how will AI impact the ecosystem and community developing future wireless technology? We trace this transformation across the entire lifecycle, from education and core research to technical publication and production-ready network deployments. As AI increasingly automates routine tasks, the primary value of the human researcher will shift from problem-solving to problem-finding, research orchestration, and oversight of trade-off management and scientific rigor. By actively preserving spaces for deep, unplugged thinking and steering AI toward genuine discovery rather than mere recombination, we can navigate this profound shift to ultimately elevate human ingenuity and the future evolution of the researcher.

cy and inference speed of AI models are expected to improve dramatically in the next few years, which will make frontier AI widely accessible at reduced cost [1].

In light of these developments, it is natural to wonder about the future role of humans in research, education, and engineering. What are the defining traits of successful researchers of the future? What should we teach, and how? Which skills will remain relevant, and which new competencies will be required? How far can we automate the research process, and how should we manage AI-generated inventions? Where will we find professional fulfillment and purpose if AI handles the heavy lifting? These are fundamental questions without clear answers. In this article, we focus on wireless communications and offer perspectives and predictions drawn from our experience in academia and industry. This article is not about how AI can enhance wireless systems, but rather how AI will impact the stakeholders devel-